Shivam Govind Rao

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Summary

AI Engineer with 2.5 years building production AI products end to end: LLM pipelines, AI agents (LangGraph, LangChain), RAG, real-time voice agents, and the Python, Node.js and React systems around them. Built the document-AI platform at Diagna AI that processes 1,000+ medical faxes/day for live US clinics, and shipped VoiceGenie, a generative-AI voice platform, from 0 to $10K MRR in 11 months. Focused on AI that works reliably in production: guardrails in code, evals, and measured latency and cost.

Experience

Software Development Engineer **Jan 2025 - Apr 2026**
Diagna AI | *diagna.ai* *Gurugram, India*

- Led multi-model AI integration on AWS Bedrock (Claude Sonnet, Amazon Nova Pro, Llama) to classify and extract structured data from **1,000+ medical faxes/day** across 40+ document categories for live paying US clinics, with clinic-specific prompts.
- Engineered LLM reliability: pre-flight token counting, context and cost aware model routing, schema-validated JSON with retries for malformed output, and a fallback chain across model families; image compression cut payloads **up to 80%**.
- Cut manual fax triage from **about 3 hours to 10 minutes a day** and cleared a recurring 2-day backlog by chaining AI extraction, patient matching, human review and EHR write-back into one workflow.
- Architected the core backend of a HIPAA-compliant platform: **40+ REST endpoints** (JWT + RBAC) over a 25-model PostgreSQL schema, and S3, SQS and SNS ingestion into **7 BullMQ/Redis queues** with idempotency, rate limiting and a dead-letter queue.
- Developed a Python FastAPI service (Robocorp, Playwright, Redis/RQ) that writes to a no-API legacy EHR plus Medplum FHIR sync; referral turnaround fell from **4.2 days to same day**.
- Delivered both production React 19 frontends, with an editor for reviewing AI output; unclear patient matches go to review.

Software Engineer **Dec 2023 - Jan 2025**
Oriserve | *voicegenie.ai* *Noida, India*

- Shipped the entire frontend of VoiceGenie, a generative-AI voice agent platform (Next.js, React, TypeScript), as the sole frontend engineer; it reached **$10K MRR in 11 months** and closed its first enterprise customers.
- Created per-campaign ElevenLabs voice controls and a runtime script composer that personalized every AI call; delivered post-call AI analytics (transcripts, emotion, lead extraction) synced to HubSpot, GoHighLevel and Cal.com.
- Integrated Stripe payments end to end; collaborated on Twilio voice-AI infrastructure for **2K+ concurrent calls** and rate limiting that blocked **15K+ abusive requests/day**; cut page load time by **30%**.

Projects

Hinglish Clinic Voice Agent (In Progress) | GitHub *Python, LiveKit Agents, Postgres | Oct 2026 - Present*

- Real-time voice receptionist that books, moves and cancels clinic appointments in English and Hinglish: LiveKit Agents, Deepgram STT, Gemini Flash, Cartesia TTS, Sarvam fallbacks, Silero VAD, barge-in.
- Safety in code, not the prompt: one agent with per-mode tool allowlists, caller verification before any personal data, read-backs built from the stored row, idempotent writes and a DB constraint against double booking.
- Evals and measurement: sandboxed multi-turn evals (**13/13 tasks, 0 wrong actions**), 217 tests, per-stage p50/p95 latency telemetry; doctor-name recognition on phone audio **61% to 99%**; slot lookup up to 1.2 s to **200 ms**.

PayOps AI: Multi-Agent Payment Reconciliation | Live Demo | GitHub *TypeScript, React 19, LangGraph | 2026*

- 19-node LangGraph system: a planner and 3 parallel agents across 5 payment systems, 15 tools, checkpointed memory, approvals.
- "Models propose, code decides": Gemini via LangChain must cite evidence, a closed 9-action catalog and policy engine bound every action, with per-run cost caps and a hash-chained audit log.
- Record/replay eval harness: **11/11 golden scenarios**, 100% match on root cause, action and policy, **$0.004** total cost; 460 tests.

ClinRAG: Hybrid RAG over FDA Drug Labels | GitHub *Python, AWS Bedrock, FAISS, RAGAS | 2026*

- Table-aware chunking, Cohere embeddings, FAISS plus BM25 with rank fusion, Claude Haiku 4.5; RAGAS on 12 hand-verified questions: context recall **0.75 to 0.83**, faithfulness 0.98 to 0.88, so reranking was skipped.

Technical Skills

AI & Agents: LangGraph, LangChain, Multi-Agent Systems, Tool Calling, RAG (FAISS, BM25), Bedrock, Claude, Gemini, OpenAI
Evals & Reliability: RAGAS, Golden Datasets, Record/Replay Evals, Guardrails, Human-in-the-Loop, Model Fallbacks
Voice AI: LiveKit Agents, Deepgram, Cartesia, ElevenLabs, Twilio, Silero VAD
Languages & Backend: Python, TypeScript, SQL, FastAPI, Node.js, Express, PostgreSQL, Redis, BullMQ, REST APIs, Pydantic
Frontend & Cloud: React 19, Next.js, Tailwind CSS, AWS (S3, SQS, SNS, Lambda), Docker, Git, pytest, Jest

Education

Institute of Engineering & Technology (IET), Dr. Ram Manohar Lohia Awadh University **2019 - 2023**
Bachelor of Technology in Computer Science and Engineering, 81% *Ayodhya, India*